Customer Behavior Project Report

Abstract
This project analyzes customer shopping behavior using Python for data cleaning, SQL for business analysis, and Power BI for dashboard visualization. The objective is to identify purchasing trends, customer preferences, and business opportunities to improve sales and customer engagement.

Introduction
The retail industry generates large volumes of customer transaction data. Analyzing this data helps businesses understand customer preferences, optimize marketing strategies, and improve decision-making.

Problem Statement
The company wants to understand consumer shopping behavior across demographics, product categories, sales channels, and seasonal patterns to improve customer satisfaction and long-term business performance.

Objectives

1. Analyze customer purchasing patterns.

2. Identify top-performing product categories.

3. Study the impact of discounts on sales.

4. Compare online and offline sales performance.

5. Build an interactive Power BI dashboard.

Dataset Description
The dataset contains customer transaction records including customer demographics, purchase amount, product category, discount percentage, payment method, sales channel, and customer ratings.

Tools & Technologies

- Python (Pandas, NumPy) – Data Cleaning

- SQL – Business Query Analysis

- Power BI – Dashboard Visualization

- Jupyter Notebook – Development Environment

Data Cleaning & Preprocessing

- Imported the dataset into Python using the Pandas library.
- Loaded the dataset into a DataFrame for analysis and preprocessing.
- Examined the initial structure of the dataset using **df.head()** to preview the first few records.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	Yes	14	Venmo	Fortnightly
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	Yes	2	Cash	Fortnightly
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	Yes	23	Credit Card	Weekly
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	Yes	49	PayPal	Weekly
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	Yes	31	PayPal	Annually

- Used df.describe() to analyze statistical summaries such as mean, median, minimum, maximum, and standard deviation.

	Customer ID	Age	Gender	Item Purchased	Category	Purchase Amount (USD)	Location	Size	Color	Season	Review Rating	Subscription Status	Shipping Type	Discount Applied	Promo Code Used	Previous Purchases	Payment Method	Frequency of Purchases
count	3900.000000	3900.000000	3900	3900	3900	3900.000000	3900	3900	3900	3900	3863.000000	3900	3900	3900	3900	3900.000000	3900	3900
unique	NaN	NaN	2	25	4	NaN	50	4	25	4	NaN	2	6	2	2	NaN	6	7
top	NaN	NaN	Male	Blouse	Clothing	NaN	Montana	M	Olive	Spring	NaN	No	Free Shipping	No	No	NaN	PayPal	Every 3 Months
freq	NaN	NaN	2652	171	1737	NaN	96	1755	177	999	NaN	2847	675	2223	2223	NaN	677	584
mean	1950.500000	44.068462	NaN	NaN	NaN	59.764359	NaN	NaN	NaN	NaN	3.750065	NaN	NaN	NaN	NaN	25.351538	NaN	NaN
std	1125.977353	15.207589	NaN	NaN	NaN	23.685392	NaN	NaN	NaN	NaN	0.716983	NaN	NaN	NaN	NaN	14.447125	NaN	NaN
min	1.000000	18.000000	NaN	NaN	NaN	20.000000	NaN	NaN	NaN	NaN	2.500000	NaN	NaN	NaN	NaN	1.000000	NaN	NaN
25%	975.750000	31.000000	NaN	NaN	NaN	39.000000	NaN	NaN	NaN	NaN	3.100000	NaN	NaN	NaN	NaN	13.000000	NaN	NaN
50%	1950.500000	44.000000	NaN	NaN	NaN	60.000000	NaN	NaN	NaN	NaN	3.800000	NaN	NaN	NaN	NaN	25.000000	NaN	NaN
75%	2925.250000	57.000000	NaN	NaN	NaN	81.000000	NaN	NaN	NaN	NaN	4.400000	NaN	NaN	NaN	NaN	38.000000	NaN	NaN
max	3900.000000	70.000000	NaN	NaN	NaN	100.000000	NaN	NaN	NaN	NaN	5.000000	NaN	NaN	NaN	NaN	50.000000	NaN	NaN

```
df.isnull().sum()
```

- Checked for missing/null values using df.isnull().sum() And

```
Customer ID                 0
Age                         0
Gender                      0
Item Purchased              0
Category                    0
Purchase Amount (USD)       0
Location                    0
Size                        0
Color                       0
Season                      0
Review Rating              37
Subscription Status         0
Shipping Type               0
Discount Applied            0
Promo Code Used             0
Previous Purchases          0
Payment Method              0
Frequency of Purchases      0
dtype: int64
```

- Handled missing values in the Review Rating column by replacing null values with the median rating
- Examined existing column names using df.columns.
- Standardized column names by converting them to lowercase and replacing spaces with underscores for better consistency and easier querying.

```
Index(['customer_id', 'age', 'gender', 'item_purchased', 'category',
       'purchase_amount_(usd)', 'location', 'size', 'color', 'season',
       'review_rating', 'subscription_status', 'shipping_type',
       'discount_applied', 'promo_code_used', 'previous_purchases',
       'payment_method', 'frequency_of_purchases'],
      dtype='str')
```

Generate

- Created customer age categories using pd.qcut() to segment customers into groups:
    - Young Adults
    - Adults
    - Middle-aged
    - Seniors

	age	age_group
0	55	Middle-aged
1	19	Young Adults
2	50	Middle-aged
3	21	Young Adults
4	45	Middle-aged
5	46	Middle-aged
6	63	Seniors
7	27	Young Adults
8	26	Young Adults
9	57	Middle-aged

- Converted categorical purchase frequency values into numerical day-based values using a mapping dictionary.
- Created a new column named purchase_frequency_days using the mapped numerical values.
- Validated the transformed purchase frequency data

	purchase_frequency_days	frequency_of_purchases
0	14.0	Fortnightly
1	14.0	Fortnightly
2	7.0	Weekly
3	7.0	Weekly
4	365.0	Annually

- Compared the discount_applied and promo_code_used columns to check whether both columns contained identical values.
- Identified redundancy between the two columns and removed the unnecessary promo_code_used column.

	customer_id	age	gender	item_purchased	category	purchase_amount_(usd)	location	size	color	season	review_rating	subscription_status	shipping_type	discount_applied	previous_purchases	payment_method	frequency_of_purchases	age_group	purchase_frequency_days
0	1	55	Male	Blouse	Clothing	53	Kentucky	L	Gray	Winter	3.1	Yes	Express	Yes	14	Venmo	Fortnightly	Middle-aged	14.0
1	2	19	Male	Sweater	Clothing	64	Maine	L	Maroon	Winter	3.1	Yes	Express	Yes	2	Cash	Fortnightly	Young Adults	14.0
2	3	50	Male	Jeans	Clothing	73	Massachusetts	S	Maroon	Spring	3.1	Yes	Free Shipping	Yes	23	Credit Card	Weekly	Middle-aged	7.0
3	4	21	Male	Sandals	Footwear	90	Rhode Island	M	Maroon	Spring	3.5	Yes	Next Day Air	Yes	49	PayPal	Weekly	Young Adults	7.0
4	5	45	Male	Blouse	Clothing	49	Oregon	M	Turquoise	Spring	2.7	Yes	Free Shipping	Yes	31	PayPal	Annually	Middle-aged	365.0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
3895	3896	40	Female	Hoodie	Clothing	28	Virginia	L	Turquoise	Summer	4.2	No	2-Day Shipping	No	32	Venmo	Weekly	Adults	7.0
3896	3897	52	Female	Backpack	Accessories	49	Iowa	L	White	Spring	4.5	No	Store Pickup	No	41	Bank Transfer	Bi-Weekly	Middle-aged	NaN
3897	3898	46	Female	Belt	Accessories	33	New Jersey	L	Green	Spring	2.9	No	Standard	No	24	Venmo	Quarterly	Middle-aged	90.0
3898	3899	44	Female	Shoes	Footwear	77	Minnesota	S	Brown	Summer	3.8	No	Express	No	24	Venmo	Weekly	Adults	7.0
3899	3900	52	Female	Handbag	Accessories	81	California	M	Beige	Spring	3.1	No	Store Pickup	No	33	Venmo	Quarterly	Middle-aged	90.0

3900 rows × 19 columns

SQL Analysis

Business Questions

- Which customers used a discount but still spent more than the average purchase amount?

gender	revenue
Male	157890
Female	75191

- Which customers used a discount but still spent more than the average purchase amount?

customer_id	purchase_amount_(usd)
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62
24	88
29	94
32	79
33	67
35	91
37	69
40	60
41	76

- Which are the top 5 products with the highest average review rating?

item_purchased	Average Review Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.8
Skirt	3.79

- Compare the average Purchase Amounts between Standard and Express Shipping.

shipping_type	round(avg(`purchase_amount_(usd)`),2)
Express	60.48
Standard	58.46

- Do subscribed customers spend more? Compare average spend and total revenue

subscription_status	total_customers	avg_spend	total_revenue
Yes	1053	59.49	62645
No	2847	59.87	170436

- Which 5 products have the highest percentage of purchases with discounts applied?

item_purchased	discount_rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

- Segment customers into New, Returning, and Loyal based on their total number of previous purchases, and show the count of each segment.

customer_segment	number of customers
loyal	3116
returning	701
new	83

- What are the top 3 most purchased products within each category?

item_rank	category	item_purchased	total_orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

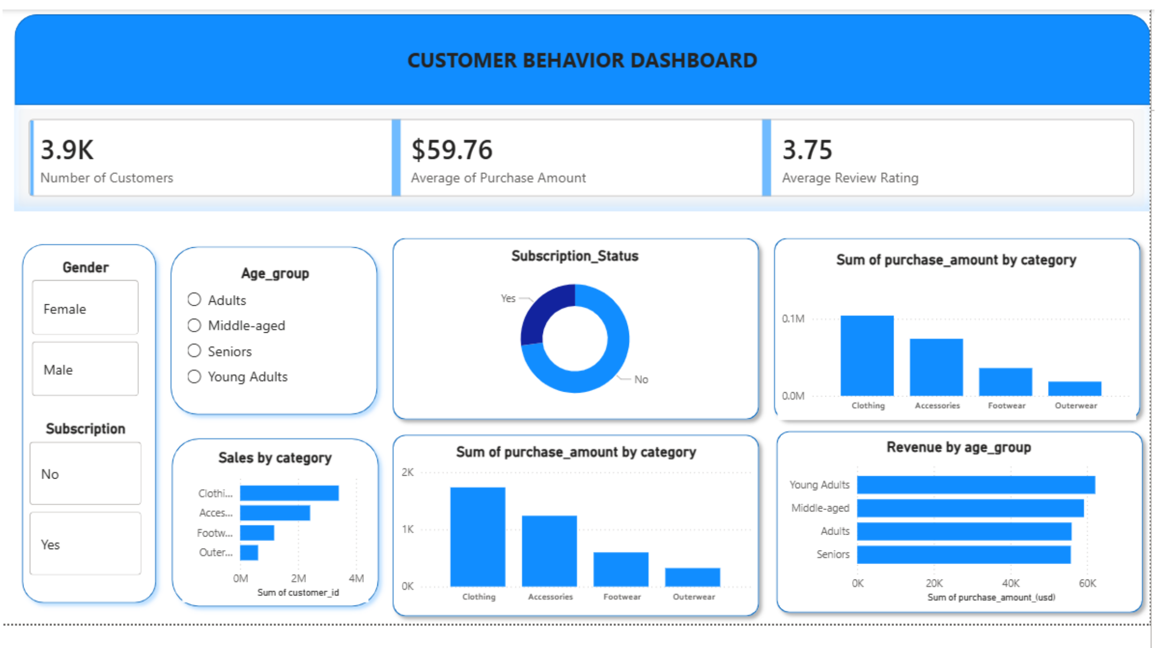
- Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?

subscription_status	repeat_buyers
Yes	958
No	2518

- What is the revenue contribution of each age group?

age_group	total_revenue
Young Adults	62143
Middle-aged	59197
Adults	55978
Seniors	55763

Dashboard Development



The Power BI dashboard includes:

- KPI Cards

The dashboard includes key performance indicators (KPIs) to provide a quick overview of customer behavior and purchasing trends:

- Total Number of Customers: 3.9K
- Average Purchase Amount: $59.76
- Average Review Rating: 3.75

- Filters and Slicers

The dashboard contains interactive filters that allow users to dynamically explore the data:

- Gender Filters
  - Male
  - Female
- Age Group filter
  - Young Adults
  - Adults
  - Middle-aged
  - Seniors
- Subscription Status filter
  - Yes
  - No

These slicers improve interactivity and enable customized analysis based on customer segments.

Visualizations Included

- Subscription Status Distribution

- Donut chart displaying the proportion of subscribed and non-subscribed customers.
- Helps identify customer subscription patterns and engagement levels.

- Purchase Amount by Category

- **Bar chart showing total purchase amount across product categories:**
  - **Clothing**
  - **Accessories**
  - **Footwear**
  - **Outerwear**
- **Used to identify high-revenue product categories.**

- Sales by Category

- **Horizontal bar chart representing customer sales distribution across categories.**
- **Helps compare category-level customer purchasing behavior.**

- Revenue by Age Group

- **Horizontal bar chart displaying revenue contribution by different customer age groups.**
- **Helps analyze which demographic group contributes the highest revenue.**

- Dashboard Features

- **Interactive filtering and cross-visual analysis.**
- **User-friendly layout for easy interpretation.**
- **Comparative analysis across customer demographics and product categories.**
- **KPI-focused design for quick business insights.**
- **Visual representation of customer purchasing behavior and revenue trends.**

Key Insights

- Clothing category generated the highest purchase amount among all product categories.
- Young Adults contributed the highest revenue compared to other age groups.
- A significant portion of customers were non-subscribers.
- Accessories and Footwear categories showed moderate sales performance.

- Customer purchasing behavior varied across demographic segments.

Recommendations

- Focus more on the Clothing category since it generates the highest revenue.
- Improve sales in low-performing categories through discounts and promotions.
- Target Young Adults with personalized marketing campaigns and offers.
- Increase customer subscriptions by providing loyalty rewards and membership benefits.
- Analyze customer reviews to improve product quality and customer satisfaction.
- Use dashboard insights regularly to support better business decisions and sales strategies.

Conclusion

This project successfully analyzed customer purchasing behavior using Python, SQL, and Power BI. Python was used for data cleaning and preprocessing, SQL was used to answer business-related questions and extract insights, and Power BI was used to create an interactive dashboard for visualization. The analysis identified key trends in customer demographics, product categories, subscription status, and purchasing patterns. The dashboard provided clear insights that can support better business decisions, improve marketing strategies, and enhance overall customer engagement.